

pation in oil and gas.

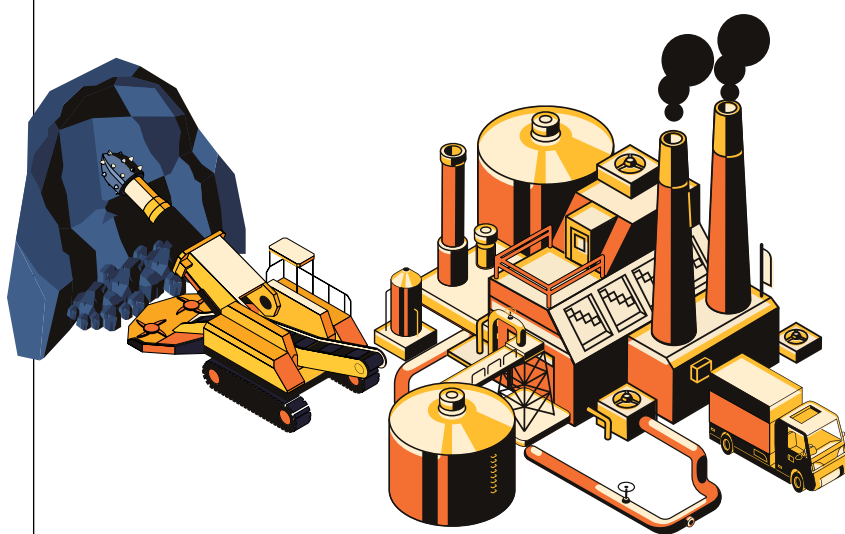
India has also ventured into overseas E&P through arms of PSUs such as ONGC Videsh, but its last major acquisition was almost a decade ago. ONGC Videsh operates 32 oil and gas projects in 15 countries, including Russia, Vietnam, Brazil, Colombia, Sudan, and Venezuela. It produced 10.278 million tonnes of oil equivalent (MTOE) of oil & gas in FY25, which is 25.3% and 15.9%, respectively, of India's domestic production.

ONGC, GAIL, and other PSUs in the sector are exploring alternative energy sources, including renewables. ONGC Green, a subsidiary of ONGC, has set a target of 10 GW renewable capacity by 2030 through solutions in green hydrogen, compressed biogas, solar, and wind.

TECHNOLOGY DEPENDENCE

India's upstream players rely heavily on foreign companies and oil majors for cutting-edge technology. ONGC and OIL have been collaborating with BP, Exxon-Mobil, Petrobras, Schlumberger, Halliburton, and Baker Hughes to overcome deep-water, seismic, and recovery challenges.

MK Sharma, former Director (Exploration & Development) of Oil India, says near-field exploration in petroleum mining lease areas for deeper horizons, as well as surrounding areas in Category-I basins, is the low-hanging fruit waiting to be plucked. "Deep-water and ultra-deep-water exploration require advanced technologies, necessitating collaboration with international agencies," he says. Indian IT firms HCLTech, Infosys, Wipro, TCS, and niche firms such as Questlabs and Nexevo are providing digital solutions. According to ONGC's digital road map submitted to the Oil Industry Safety Directorate, Infosys and Tata Consultancy Services now bundle data analytics suites, IoT sensor grids, and edge-computing gateways into off-the-shelf "platforms for petroleum," slashing implementation costs by up to 40% compared with global service companies. Still, the oil majors are heavily dependent on foreign oilfield service firms for



IS COAL GASIFICATION ON THE WAY OUT?

► India holds the fifth-largest coal reserves in the world—401 billion tonnes as of April 2025. To tap this abundant resource and meet its clean energy goals, the government is working on coal gasification, an alternative and more sustainable option. It launched the National Coal Gasification Mission in 2020 to gasify 100 million tonnes of coal by 2030.

Since then, seven coal gasification projects, mainly by PSUs, have been approved, of which three are already under implementation. The commissioning target is FY30. A total ₹6,133.06 crore has been earmarked to support these projects. China is already a big player in coal gasification. In 2024, 276 million tonnes of coal—equal to the annual consumption of Europe—was converted to chemicals, oil, and natural gas, according to the China National Petroleum and Chemical Planning Institute. The focus on coal gasification helped the country hedge the oil and gas supply shock posed due to closure of the Strait of Hormuz.

In 2024, the government had approved an outlay of ₹8,500 crore for the promotion of coal/lignite gasification projects for both PSUs and the private sector.

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hardware integration.

Another hurdle is availability of rigs. Indian PSUs hire rigs for short-term contracts of up to two years, leading to shortages when demand increases. ONGC and Oil India have some rigs that are ageing. Indian explorers are finding it difficult to source rigs, especially jack-up, with a jump in exploration activity globally following the Russia-Ukraine war. ONGC has issued a \$20 billion tender for hiring rigs for five years as part of the government's Samudra Manthan Mission. The tender, floated earlier this February, sets an 80-day mobilisation timeline, an indication of the urgency to scale exploration.